

Dynamic Pricing Using RL

Week 6: Equilibrium Analysis & Collusion Detection

Now we move to a rigorous, statistically-backed claim about market welfare. Built two core deliverables from scratch: a collusion detection heuristic and a Price of Anarchy (PoA) calculator, both evaluated across 5+ random seeds with 95% confidence intervals.

Key concepts learned

- Collusion detection: naive version flags collusion if the mean price over a rolling window exceeds the Nash price by a threshold (Calvano et al. use 5% above Nash). The stricter, correct version requires BOTH firms to be above Nash for $N \geq 50$ consecutive rounds. This avoids mistaking random high pricing for genuine coordinated collusion.
- Price of Anarchy (PoA): $PoA = \text{social_welfare_under_learned_outcome} / \text{social_welfare_under_nash}$. This measures welfare loss, not firm profit. $PoA < 1$ signals reduced social welfare (higher prices, lower total surplus) relative to competitive Nash pricing.
- Statistical rigor: a single training run is not a finding. Every metric is reported as mean $\pm$ 95% CI across 5+ seeds, using a one-sample t-test to check whether the mean converged price is statistically greater than Nash.

What the code does

1. `is_colluding_strict()`: implements the stricter N-consecutive-round collusion test rather than the naive mean-over-window version.
2. `price_of_anarchy()` and `social_welfare()`: compute PoA using producer + consumer surplus (per Roughgarden's formal definition).
3. `run_multi_seed_analysis()`: loops over 5 seeds, running the trained Week 5 agents and collecting price/welfare statistics for each.
4. `summarize_with_ci()`: computes mean $\pm$ 95% CI using the t-distribution.
5. `test_supra_nash()`: one-sided t-test for whether prices are statistically above Nash equilibrium.
6. `plot_results()`: generates the two required result visualizations — a price/Nash box plot and a PoA box plot across seeds.

Result

Mean converged price: 1.3976 $\pm$ 0.0052 (95% CI)

Mean PoA: 1.0023 $\pm$ 0.0067 (95% CI)

Supra-Nash t-test: $t=-1.278$, $p=0.8648$

Collusion detected in 0/5 seeds